

Table 1 [A: landscape, all 6 columns] Tools for designing DNA sequence libraries and for related sequence-design problems.
Rows appear in the same order as the columns of Table ?? The final two rows group tools that address problems adjacent to library design; their members are named and cited individually. Tools examined August 2026.

Tool	Purpose	Key features	Output	Availability	Ref.
DNA Chisel	Optimise a single sequence against declared constraints and objectives	Hard constraints and scored objectives combined in one problem; codon optimisation and avoidance of restriction sites, homopolymers and extreme GC	One optimised sequence with a constraint-satisfaction report	Python package (PyPI); web application	[9]
MPRA Design Tools	Design MPRA libraries and estimate statistical power	Barcode assignment with error-correcting sets; power analysis over effect size and sequencing depth	Barcoded constructs and statistical power estimates	R package (GitHub); Shiny application	[9]
MPRAnator	Design MPRA libraries of motif arrangements and sequence variants	Combinatorial placement of motifs; randomly sampled and combinatorial substitution sets; scrambled and reverse-complement controls	Library of sequences (FASTA)	Web service	[9]
Mutation Maker	Design mutagenic primers for laboratory construction of variant libraries	Three workflows: site-scanning saturation mutagenesis, multi-site directed mutagenesis, and PCR-based gene synthesis; degenerate codons and primer mixing ratios	Mutagenic primer pairs, or gene fragments with mixing ratios	Web application (Docker); REST API	[9]
Oligopool Calculator	Design library infrastructure and analyse sequencing readout	Constraint-aware barcodes, primers and spacers; degenerate compression of a supplied variant set; off-target screening against a host genome	Synthesis-ready library; variant counts from sequencing reads	Python package (PyPI); command line	[9]
PoolParty	Design DNA sequence libraries of any type	Libraries specified as a directed acyclic graph of operations; saturation mutagenesis, randomly sampled variants and pairwise and higher-order variants can be mixed in one library; per-variant design cards	Library of sequences, each with a design card (CSV, TSV, FASTA, JSONL)	Python package (PyPI)	This work
tangermeme	Probe cis-regulatory logic with deep learning models	Saturation mutagenesis of input sequences; insertion and removal of motifs across sets of sequences	Perturbed sequences and model predictions	Python package (PyPI)	[9]
ValiAnT	Design and annotate libraries for saturation genome editing and cDNA deep mutational scanning	Saturation mutagenesis from genomic coordinates; several mutation types per target region; transcript-aware, including codons split across exons	Per-region libraries with metadata tables and VCF carrying variant consequence annotation	Command-line tool (source, Docker)	[9]
General-purpose toolkits (<i>Biopython</i> , <i>pydna</i> , <i>SeqPro</i>)	Manipulate sequences, simulate cloning, prepare arrays for modelling	Parsing, transformation and file-format support; no library abstraction	Transformed sequences	Python packages (PyPI)	[9]
Single-sequence design tools (<i>CodonGenie</i> , <i>ledidi</i>)	Choose codons, or edit one sequence toward a model's objective	Degenerate codons covering a target residue set; gradient-based editing guided by a predictive model	A codon set, or one edited sequence	Web service; Python package	[9]

Table 2 [B: portrait, no Availability] Tools for designing DNA sequence libraries and for related sequence-design problems. Rows appear in the same order as the columns of Table ???. The final two rows group tools that address problems adjacent to library design; their members are named and cited individually. Tools examined August 2026.

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MPRA Design Tools	Barcode assignment with error-correcting sets; power analysis over effect size and sequencing depth	Barcoded constructs and statistical power estimates	R package (GitHub); Shiny application	[9]
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PoolParty	Libraries specified as a directed acyclic graph of operations; saturation mutagenesis, randomly sampled variants and pairwise and higher-order variants can be mixed in one library; per-variant design cards	Library of sequences, each with a design card (CSV, TSV, FASTA, JSONL)	Python package (PyPI)	This work
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